

Ryan Gilmour

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Personal Statement

I am a software engineer with 7+ years' experience delivering software and data solutions across the energy sector. I am experienced in software design, machine learning, data engineering and energy systems modelling. I have a strong understanding of energy systems, with experience translating complex engineering problems into practical and scalable software solutions. I am a clear communicator with experience collaborating across technical teams, mentoring colleagues and presenting technical concepts to industry audiences.

Technical Skills

Software Engineering: Python, Git, CI/CD, Test Driven Development, Agile, REST APIs, Kafka

Data Engineering: SQL, Spark, Delta Lake, ETL, dimensional modelling

Cloud & Infrastructure: AWS (Glue, Lambda, Redshift, EC2, CloudFormation), Docker, Kubernetes

Employment

Jun 2024 – Current

Hiko Energy Insights – Software Engineer

Developing an energy insights platform used by eight New Zealand distribution network companies to manage their LV networks.

- Designing and building ETL pipelines using data lake and dimensional modelling techniques, including daily ingestion of smart metering data from 250,000 meters using Spark.
- Researching, designing and productionising a demand forecasting tool across all networks.
- Innovating with network engineers on novel LV fault detection methodologies using data.
- Designing and implementing APIs to support distributed generation planning use cases.
- Acting as the key power systems engineering resource for technical design questions.
- Mentoring colleagues to develop design, communication and software development skills.
- Presenting at key industry conferences including on how to Productionise Machine Learning.

Jan 2023 – May 2024

GE Vernova – Solutions Specialist

Delivered highly available DERMS grid management software to utilities internationally as part of the Services Delivery team.

- Defined agile working processes for a new 8-person Services Engineering team.
 - Led the technical delivery of a long-term vehicle-to-grid study for a major customer.
 - Trained forecasting machine learning algorithms to optimise for scalability and accuracy.
 - Developed custom Python microservices to integrate customer data with GE software.
 - Interfaced with the product team to influence development to support customer use cases.
 - Delivered workshops on renewables integration and data formats for energy modelling.
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Aug 2021 – Jan 2023

Scene Connect – Simulations Lead

Led the design and development of ZUoS Simulations, the company's core software product for the design and optimisation of Smart Local Energy Systems to reach net zero.

- Led the architectural design and product roadmap in collaboration with customers and the business steering group.
- Co-developed the product's core optimisation module alongside academic researchers.
- Developed a probabilistic Monte Carlo simulation module and machine learning models for energy demand forecasting.
- Led the technical delivery of projects with Distribution System Operators assessing the impact of heat pumps, EVs and battery storage on electricity networks.
- Created digital twins of distribution networks using GIS and network planning data.
- Provided technical demonstrations and consultancy support to network operators, generators and energy modelling consultants.

Aug 2019 – Aug 2021

Scene Connect – Electrical Engineer

Supported low-carbon transition projects through energy consultancy and R&D.

- Performed grid connection studies and technical analysis for renewable energy projects.
- Developed geospatial analyses and feasibility studies for low-carbon technologies.
- Supported community energy projects and innovative R&D programmes.

Education

2014 – 2019

The University of Edinburgh

M.Eng Electrical Engineering with Renewable Energy — 1st Class

Volunteering

Code Club Aotearoa (2025)

- Teaching primary school children basic coding skills.

Scouts & Camp America (2011-2014; 2015; 2018)

- Teaching young people outdoor skills.

Engineering for Change (2018)

- President of a sustainable engineering student society.

Hobbies

Rock climbing, camping, cooking, travelling.